

| Assessment | Test | Domain        | Skill               | Difficulty                                          |
|------------|------|---------------|---------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Medium <div><div></div><div></div><div></div></div> |

Question

$q(x) = 32(2^x)$

Which table gives three values of  $x$  and their corresponding values of  $q(x)$  for function  $q$ ?

Answer

- A.

|        |       |     |      |
|--------|-------|-----|------|
| $x$    | $-1$  | $0$ | $1$  |
| $q(x)$ | $-64$ | $0$ | $64$ |
- B.

|        |                |     |      |
|--------|----------------|-----|------|
| $x$    | $-1$           | $0$ | $1$  |
| $q(x)$ | $\frac{1}{16}$ | $2$ | $64$ |
- C.

|        |                |      |      |
|--------|----------------|------|------|
| $x$    | $-1$           | $0$  | $1$  |
| $q(x)$ | $\frac{1}{16}$ | $32$ | $64$ |
- D.

|        |      |      |      |
|--------|------|------|------|
| $x$    | $-1$ | $0$  | $1$  |
| $q(x)$ | $16$ | $32$ | $64$ |

Correct Answer: D

Rationale

Choice D is correct. Substituting  $-1$  for  $x$  in the given function yields  $q(-1) = 32(2)^{-1}$ , which is equivalent to  $q(-1) = 32(\frac{1}{2})$ , or  $q(-1) = 16$ . Therefore, when  $x = -1$ , the corresponding value of  $q(x)$  for function  $q$  is  $16$ . Substituting  $0$  for  $x$  in the given function yields  $q(0) = 32(2)^0$ , which is equivalent to  $q(0) = 32(1)$ , or  $q(0) = 32$ . Therefore, when  $x = 0$ , the corresponding value of  $q(x)$  for function  $q$  is  $32$ . Substituting  $1$  for  $x$  in the given function yields  $q(1) = 32(2)^1$ , which is equivalent to  $q(1) = 32(2)$ , or  $q(1) = 64$ . Therefore, when  $x = 1$ , the corresponding value of  $q(x)$  for function  $q$  is  $64$ . Of the choices given, only the table in choice D gives these three values of  $x$  and their corresponding values of  $q(x)$  for function  $q$ .

Choice A is incorrect. This table gives three values of  $x$  and their corresponding values of  $q(x)$  for the function  $q(x) = 32(2x)$ .

Choice B is incorrect. This table gives three values of  $x$  and their corresponding values of  $q(x)$  for the function  $q(x) = 2(32)^x$ .

Choice C is incorrect and may result from conceptual or calculation errors.